

Subhash BM

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Staff Research Scientist with over 9+ years of experience building **large-scale distributed STOA ML systems** and applications. Proven ability to architect high-performance **end to end ML pipelines**, across diverse technology stacks, optimize performance, and mentor engineering teams.

EXPERIENCE

Meta Platform

Bellevue, WA

Staff Research Scientist

Jan 2026 - Present

- Designed and built the Feature Space Optimization Platform (FSOP), an end-to-end automated system that orchestrates distributed ML experiments across 10-20 production models simultaneously on Meta's ML Pipeline Platform (MLPP), enabling feature evaluation, model retraining, and NE analysis at petabyte scale.
- Achieved 30-60% feature space reduction across 5 major head/neck advertising ranking models in MC2 2025, removing **5,624 features** while maintaining prediction quality (NE neutrality $\pm 0.025\%$), yielding an estimated **\$44.5 million per year** in inference compute cost savings.
- Architected the FSOP Workflow Insights dashboard consolidating 5 heterogeneous data sources (SQL warehouses, real-time event streams, graph databases, blob storage, operational databases) into a unified monitoring platform for ~20 advertising model types across hundreds of training workflows.

Microsoft Azure

Redmond, WA

Senior Machine Learning Engineer

Jul 2024 - Jan 2026

- Provided **technical leadership** through design discussions, cross-functional collaboration, and mentorship, fostering a supportive team environment and accelerating onboarding and knowledge transfer.
- Working with **cross-functional teams** (Engineering, PM, Marketing, Field, Finance, Leadership) to build Machine learning solutions and running product led growth campaigns for customer life cycle management for M365 customers and partners.
- End-to-end development of **Risk Classification Models** based Neural Networks and deep Learning applications for millions of users using **Pytorch**.
- Build end-to-end training pipeline using **ML flow**, including data sampling, HRS labeling, **feature extraction**, model training, offline evaluation, online experiment, flight tuning and analysis.

Technologies:: C# .NET, Aspire, Cosmos DB, Azure Pipelines, GraphQL, Pytorch, Kubernetes, ML flow

Machine Learning scientist II

Jan 2023 - Jul 2024

- Optimized request handling by categorizing and filtering requests to downstream services, reducing failures due to node repairs and improving system reliability.

Skills :: C#, TensorFlow, PyTorch, SparkML, python, R, Neural Network(CNN), Bert, Data mining

Amazon Web Services

Seattle, WA

Software Dev Engineer II

Dec 2021 - Dec 2022

- Engineered and led the modernization of the core service registry platform to a high-performance, Java-based stack (Spring Boot, GraphQL, gRPC), successfully **eliminating substantial technical debt** and bolstering scalability for distributed services operating across complex multi-cloud environments.
- Architected and deployed a new, fully automated CI/CD pipeline leveraging AWS CDK, which featured integrated rollbacks and staged deployments, reducing critical deployment times to 4 hours. Co-led the design and development of highly scalable Dataplane cells, enabling ingestion and processing of **Petabyte-scale streaming time-series data** and boosting overall **system efficiency by 35%**.
- Drove a **40% improvement in operational excellence** by designing and implementing advanced CloudWatch monitoring and alarming systems for EKS clusters using **AWS CDK**, ensuring high availability for critical production systems,
- Contributed key performance enhancements to the **open-source CNCF Cortex project**.

Technologies: Go, Java (Spring Boot), GraphQL, gRPC, Kubernetes, Docker, Istio, Envoy, Linkerd, AWS (CDK, EKS, CloudWatch, SDK), React.js, Typescript, Node.js, SQL, Linux.

Octagon Technologies

Bangalore, India

Software Engineer

Feb 17 - Jul 21

- Architected and implemented scalable data persistence layers using Spring Boot and JDBC with MySQL, focusing on efficient ORM and optimized data models to ensure **high-performance storage and retrieval** in enterprise applications.
- Designed and deployed robust RESTful APIs utilizing Spring Boot, adhering to established architectural patterns and best practices for clear endpoint definitions, optimized JSON serialization, and seamless, high-throughput client-server communication.

Technologies: Spring Boot, MySQL, JDBC, Log4j, Maven, Apache Tomcat, RESTful, basic SQL optimization techniques.

EDUCATION

State University of New York At Buffalo

Buffalo, NY

Master of Science in Data Science - Specializing in Distributed systems and Machine learning

Research Scientist *City University of New York, NY*

- Researched carbon emissions and sinks in global urban centers using Sentinel satellite imagery and advanced ETL/image classification workflows.
- Built a multimodal news image ranking recommendation model with TensorFlow and PyTorch, leveraging collaborative filtering and data analysis to optimize content relevance.